

The Conditioned Covered Call

A volatility-responsive NIFTY call overlay on Model A and Model D

A weekly short NIFTY call is written against the equity book. Unlike a conventional covered call, the strike is not fixed at a constant distance — it slides with how volatile the market has actually been. When NIFTY is quiet the strike is pulled in below spot; when NIFTY is turbulent it is pushed well above. The result is a leg that behaves as a variable delta hedge rather than a premium-harvesting scheme.

$$K = \text{Spot} - 150 + 200 \times z$$

z = 14-day realised volatility, z -scored on its own trailing 252 sessions, clipped ± 3

Model A, return on Rs 1 Cr	20.19% → 25.00%	Model D, return on Rs 1 Cr	17.21% → 22.02%
Model A, max drawdown	−24.06% → −17.70%	Model D, max drawdown	−17.51% → −13.64%
Test window	Feb 2021 – Jul 2026	Weekly cycles / transactions	277 / 744
Walk-forward OOS (honest)	25.69% / 18.83%	Placebo p-value	0.048

Rs 1,00,00,000 fixed capital, pledged for F&O margin · profit booked monthly, no compounding · real NSE lot sizes · five-minute option data

1. Executive Summary

Model A and Model D are equity portfolios that deliver roughly 20% and 17% a year on a fixed Rs 1 crore, with drawdowns of 24% and 18% respectively. This note documents an options overlay that raises both returns and lowers both drawdowns, and — more importantly — documents the tests run to establish that the improvement is not an artefact of curve-fitting.

Rs 1,00,00,000 deployed, profit booked monthly, no compounding. Slippage 0.25 pts/leg.

Book	Return %/yr	Max DD %	Vol %	Sharpe	CAR/MDD	Worst month	Win months %
Model A only	20.19	-24.06	17.41	0.82	0.84	-12.02	65.2
Model A + current 2.5 σ call	20.74	-23.73	17.30	0.86	0.87	-11.94	66.7
Model A + conditioned	25.00	-17.70	14.11	1.35	1.41	-9.69	65.2
Model D only	17.21	-17.51	15.68	0.72	0.98	-11.66	66.7
Model D + current 2.5 σ call	17.76	-17.13	15.54	0.76	1.04	-11.59	66.7
Model D + conditioned	22.02	-13.64	12.33	1.31	1.61	-9.33	69.7

Reading note: "Return %/yr" is *simple* return — total profit divided by years divided by Rs 1 crore — not CAGR, because the mandate withdraws profit monthly and nothing compounds. The CAGR-equivalent for Model A + conditioned is 17.07%; quoting 25.00% against another strategy's CAGR would not be a fair comparison. Sharpe is computed on monthly returns, which understates volatility slightly less than a daily calculation would (the option leg settles as a lump on expiry day); the daily-basis Sharpe is 1.10 for Model A and 1.02 for Model D. Full detail in §9.4.

Findings

- **The overlay contributes Rs 26,31,511** over the period — 19% of Model A's total profit and 22% of Model D's — and is positive in 59% of months.
- **It earns most when the models struggle.** In 2026 it turned Model D from -3.15% into +5.24%. In 2021, the strongest equity year, it cost 2.21 points.
- **94% of the advantage comes from 33 violent weeks** out of 277. In those weeks a fixed-strike call loses money; this one has already moved out of range.
- **It survives walk-forward.** Choosing parameters using only prior data gives 25.69% (Model A) and 18.83% (Model D) for 2023–2026, against 20.18% and 13.32% unhedged. Roughly one percentage point of the headline is hindsight; about 5.5 points is real.
- **It survives a placebo.** Run 20 times with the signal randomly shuffled, none of the 20 controls beat it and 18 lost money. Empirical $p = 0.048$.

The honest number to plan around is the walk-forward result, not the headline. The figures in the summary table use the best settings found across the whole period, which could not have been known in advance.

2. The Two Books — Model A and Model D

The overlay in this note is layered on top of two pre-existing equity strategies, run independently of each other. Both are systematic NIFTY 50 stock-selection models built from the same underlying universe and point-in-time reshuffle history, but they select stocks by different methods and carry different macro awareness. Their own standalone backtest statistics are summarised below for reference; the sections that follow build the overlay on top of each.

Model A — Fixed-Alpha Sector-Mean Lasso

Method	Sparse Lasso regression (L1-penalised, $\alpha=0.001$) selecting the stocks whose returns best replicate each sector's mean return
Universe	NIFTY 50 constituents, point-in-time historical basket (survivorship-bias free)
Rebalance	Monthly, 252-session rolling lookahead
Avg. stocks held	13.7 — a concentrated, sparse book
Own backtest CAGR	22.44%/yr vs NIFTY 50's 11.12%/yr
Own backtest Sharpe	0.91 vs NIFTY 50's 0.36
Own backtest max DD	−21.47% vs NIFTY 50's −17.23%
Correlation to NIFTY (model's own figure)	0.868
Monthly win rate vs NIFTY	62.1% (41 of 66 months)

Model D — Shrinkage QP with Macro Integration

Method	Quadratic-programming portfolio optimiser with covariance shrinkage, conditioned on four macro signals
Macro inputs	India VIX (dynamic 63-day crisis / 252-day normal lookahead), RBI repo rate, 1-year G-Sec yield (real risk-free rate), INR/USD stress flag
Rebalance	Monthly, same point-in-time universe as Model A
Avg. stocks held	24.7 — a broader, more diversified book
Own backtest CAGR	17.86%/yr vs NIFTY 50's 11.12%/yr
Own backtest Sharpe	0.73 vs NIFTY 50's 0.37
Own backtest max DD	−16.41% vs NIFTY 50's −17.23%
Correlation to NIFTY (model's own figure)	0.950
Monthly win rate vs NIFTY	69.7% (46 of 66 months)

Both backtests cover 2021-02-01 to 2026-07-22 (66 months), pass their own look-ahead and survivorship checks (100% out-of-sample rolling, point-in-time basket), and are computed on a compounding basis in their native reports — the natural convention for a standalone equity strategy. The figures quoted from §3 onward in this note use the Rs 1 crore, profit-booked-monthly, non-compounding mandate described next, plus a small additional daily cost drag (1.317% p.a. for Model A, 0.225% p.a. for Model D) to bring both onto a common cost basis with the option overlay. The two sets of numbers are not meant to match line for line — they answer different questions under different capital-treatment conventions — and both are reported in full so nothing is hidden behind a single number.

Model D carries roughly 1.8× as many holdings as Model A and is more diversified, macro-aware and closer to the index (correlation 0.950 vs 0.868). That shows up consistently through this note: Model D starts with a lower standalone return and a tighter drawdown than Model A, and the overlay's proportional improvement is larger on Model D in every comparison in §7 and §9.

3. Structure and Mandate

Two components share one Rs 1 crore.

The equity book

The full Rs 1 crore is invested in the Model A or Model D stock portfolio and **pledged** with the broker. Pledging attracts a 50% haircut, so the holding produces **Rs 50,00,000 of F&O margin** without any stock being sold. Nothing is bought on borrowed money and no separate cash sleeve is held.

The option overlay

A short NIFTY call, one weekly expiry at a time, sized so its notional matches the equity book. Average coverage is 94.8% and never exceeds 100% — the position is covered at all times and never naked.

The capital mandate

Rs 1 crore is deployed and never grows. At each month end the month's profit or loss is booked out to a separate ledger and the book is rebalanced to exactly Rs 1 crore. Returns are therefore **simple, not compounded**; no CAGR applies, and the correct figure is average annual profit as a percentage of the Rs 1 crore deployed. Drawdown is measured on the cumulative booked-profit curve and expressed against the same Rs 1 crore.

4. The Trading Rule

- 01 **Friday 15:25** — compute the 14-day realised volatility of NIFTY and its z-score against the trailing 252 sessions. This is z .
- 02 **Friday 15:25** — set $K = \text{Spot} - 150 + 200 \times z$, rounded up to the 50-point grid. If that exact strike is unquoted, take the nearest listed strike above it.
- 03 **Friday 15:25** — sell lots = $\text{floor}(\text{Rs } 1 \text{ Cr} \div (\text{entry spot} \times \text{lot size}))$, capped by the pledge margin. Sell the call for the expiry two sessions away.
- 04 **Intraweek** — if NIFTY moves 300 points either way from the strike anchor, buy the call back and re-sell at the new K , holding that week's z . Fires 0.34 times per week.
- 05 **Tuesday 15:25** — buy the position back. It is never allowed to expire or settle, which also avoids exercise STT.

Expiry is currently Tuesday, so a two-session hold means entering Friday and closing Tuesday — two trading sessions but four calendar days, capturing weekend decay. The rule adapts automatically if the expiry weekday changes, as it did from Thursday in 2024.

Worked example

NIFTY at 24,000, $z = 0$ (ordinary volatility), lot size 65. Strike = $24,000 - 150 = \mathbf{23,850}$. Lots = $\text{floor}(1,00,00,000 \div (24,000 \times 65)) = \mathbf{6}$. Margin = $6 \times 23,850 \times 11\% = \mathbf{Rs } 10,29,600$, about a fifth of the pledge. Sell six lots of the 23,850 CE for Tuesday's expiry; re-strike if NIFTY trades through 24,300 or 23,700.

5. Measuring Volatility: RV, IV and the Spread

The overlay rests on the difference between the volatility the option market *charges* and the volatility that subsequently *arrives*. Both are measured daily from the project's own data.

Realised volatility (RV)

Input	NIFTY 15:25 close each session
Daily move	$\ln(C_h / C_{h-1})$
Window	14 sessions, ending the prior day
Annualised	$\text{stdev} \times \sqrt{252}$
Median	11.3% (range 4.2–34.9%)

Implied volatility (IV)

Input	ATM call, nearest live expiry, 15:25
Approximation	$C \approx 0.4 \times S \times \sigma \times \sqrt{T}$
Inverted	$\sigma = C / (0.4 \times S \times \sqrt{T})$
T	trading days to expiry ÷ 252
Median	13.6% (ATM premium 116 pts)

The variance risk premium. Across 1,242 trading sessions, IV exceeded RV in **69.4%** of them, by an average of **+1.63 volatility points**. That persistent overcharge is the economic engine of the strategy. As a cross-check, the IV estimate (median 13.6%) sits alongside India VIX (median 14.4%) — the figure is read from the market, not modelled.

Which measure drives the strike

All four candidates were tested head to head across the same grid.

Best achievable result per signal — Model A, return on Rs 1 Cr, full window

Signal	Question it answers	Strike moves	Return %/yr
RV — realised	How much will it move?	Out, when vol is high	25.19
India VIX	How nervous is the market?	Out, when vol is high	24.86
IV – RV spread	How well am I being paid?	In, when premium is rich	24.33
IV — implied	What is being charged?	Out, when vol is high	24.06

The third row runs deliberately opposite to the others. The IV–RV spread argues that when insurance is unusually overpriced one should sell *closer* to the market, because the compensation per unit of risk is high. Realised volatility argues that when the market is genuinely thrashing one should move *further away*, because a large move is coming. Both are coherent; over five and a half years of prices, forecasting the move proved more valuable than chasing the fee. The IV–RV spread is retained as a health indicator: if the +1.63 point gap closes, the structure loses its engine.

6. Why It Works — the Mechanism

The headline premiums are misleading and it is important to be precise about this. When the rule sells a call 391 points in the money for 405 points, only about 15 of those points are *time value* — compensation collectable if spot does not move at all. The remaining 391 is intrinsic value, repaid on close. That does not mean the week's actual P&L is only 15 points: in the quiet weeks studied, NIFTY drifted an average of −36 points, and a near delta-1 short call profits when the market falls, so the realised outcome in those weeks averaged **43 points net** — 15 from time value, the rest from direction. The point of the decomposition is not that 15 points is "the" income; it is that only the time-value component is *new* money at the level of the whole book. The directional component is the hedge doing its job, and it is offset almost exactly by a matching loss on the equity side (see §9). Every figure below is reconciled to the traded rupee P&L, not to the intrinsic split, which is descriptive only.

Premium decomposition and effective hedge ratio, per lot, in index points

Market state	Weeks	Strike vs spot	Premium	Intrinsic	Time value	Approx. delta
Very quiet	43	−391	405.0	390.7	14.2	~1.00
Quiet	91	−265	291.4	265.1	26.3	~0.80
Normal	74	−124	186.0	123.8	62.2	~0.70
Busy	36	+8	105.9	17.8	88.1	~0.50
Violent	33	+256	47.6	0.0	47.6	~0.20

Read the delta column. **The rule is modulating a hedge, not harvesting premium.**

- **Quiet market** → **delta ≈ 1.0** → **hedge on**. The short call offsets the equity book's NIFTY exposure almost exactly. Market risk falls close to zero, while the portfolio's stock-selection alpha — the component that is not NIFTY — is untouched. The time value is collected on top.
- **Violent market** → **delta ≈ 0.2** → **hedge off**. The book keeps its full participation, and the far-out call collects pure time value with little chance of breach.

Average real income across all states is 44.6 points per lot per week, or about Rs 2,898.

Where the money is actually made

Per lot per week, bucketed by the signal read before the trade

Market state	Weeks	Move that followed	Conditioned strike	Fixed strike	Difference
Very quiet	43	198 pts	+2,816	+2,399	−266
Quiet	91	272 pts	+1,343	+958	+41
Normal	74	319 pts	+964	+1,116	+2
Busy	36	375 pts	+1,549	+1,297	+392
Violent	33	395 pts	+1,638	−688	+3,058

Two observations. First, **the signal forecasts**: the realised move rises monotonically with z , from 198 to 395 points. Second, **the entire advantage is in the last row**. In the other four states the conditioned and fixed rules are effectively tied. In violent weeks the fixed-strike version is run over by sharp rallies and loses money, while the conditioned version has already stepped aside. That single row is the strategy.

7. Benchmark Correlation — Independent Confirmation

Everything so far has been argued from P&L. This section tests the same claim a different way: measuring correlation directly against the NIFTY benchmark and against the equity leg itself. If the mechanism described in §6 is real, three things must be true in the correlation numbers independently of the P&L numbers — and all three are.

7.1 Model A and Model D are highly NIFTY-beta portfolios to begin with

Daily return correlation to NIFTY, 2021-02-04 to 2026-07-06 (n=1,333 sessions)

Book	Correlation	R ²	Beta	Annualised alpha
Model A	0.853	0.728	1.082	+6.89%
Model D	0.938	0.880	1.075	+3.88%

Both books move with the market almost one-for-one (beta ≈ 1.08) and NIFTY alone explains 73–88% of their daily variance. The remainder — the alpha column — is the stock-selection edge the overlay is designed to preserve rather than hedge away.

7.2 The option leg's sensitivity to NIFTY falls exactly as volatility rises

For each traded week, the option leg's P&L (points per lot) was regressed against that week's NIFTY move. The slope of that regression is an empirical estimate of the position's delta — how many points the option leg gains or loses per point NIFTY moves — measured directly from realised outcomes, independent of any option-pricing model.

Empirical delta of the option leg, by volatility regime read before the trade

Market state	Weeks	Slope (pts P&L / pt NIFTY)	Correlation
Very quiet	43	−0.980	−0.993
Quiet	91	−0.916	−0.985
Normal	74	−0.738	−0.961
Busy	36	−0.517	−0.887
Violent	33	−0.145	−0.669

In very quiet weeks the option leg tracks NIFTY almost perfectly one-for-one in the opposite direction (slope −0.98, a near-complete hedge). In violent weeks that sensitivity collapses to **−0.15**, close to unhedged. This is the delta-modulation claimed in §6, measured empirically rather than assumed — the slope falls monotonically across all five buckets, exactly as the strike-selection rule intends.

7.3 The option leg diversifies the book precisely when equity is falling

Monthly, correlation between the equity leg and the option leg

Book	All months (n=66)	Equity-down months only	Avg option leg, equity-down months	Avg option leg, equity-up months
Model A	−0.650	−0.594 (n=23)	+1.69%	−0.29%
Model D	−0.672	−0.379 (n=22)	+2.05%	−0.43%

Correlation between the two legs is negative in both the full sample and, more importantly, when isolated to **only the months the equity book lost money**. In those months the option leg earned a positive return roughly six to

seven times larger than in equity-up months. This is a direct test of the diversification claim, and it holds.

7.4 Net effect on the whole book's market correlation

Correlation of the total monthly book return to NIFTY, with and without the overlay

Book	Without overlay	With overlay	Change
Model A	+0.833	+0.734	−0.098
Model D	+0.949	+0.872	−0.077

The overlay measurably reduces market correlation on both books, by roughly a tenth of a correlation point. This is a modest shift by design — the intent is never to remove NIFTY exposure entirely (that would sacrifice the beta return the mandate wants), only to damp it exactly when volatility is elevated, which §7.2 shows it does.

Reading these four results together: the equity books are NIFTY-beta vehicles (7.1); the overlay's sensitivity to NIFTY collapses precisely when the volatility signal says a big move is likely (7.2); the overlay's return is negatively correlated with the equity leg and concentrated in the equity book's losing months (7.3); and the net effect, measured directly on the combined book, is a real but deliberately partial reduction in market correlation (7.4). Every one of these is an independent statistical test of the mechanism claimed in §6, computed from correlation coefficients rather than P&L, and all four point the same way.

8. Benchmark Comparison — Growth, Distribution and Drawdown

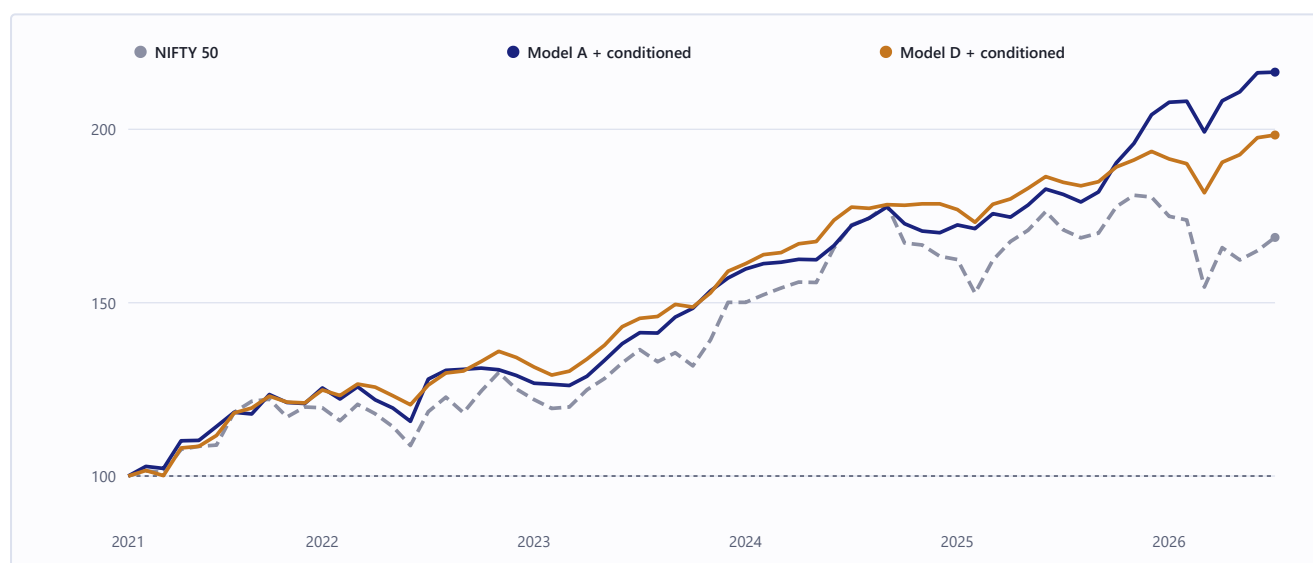
The tables in §7 established correlation statistically. This section shows the same comparison visually, against a plain buy-and-hold Rs 1 crore position in NIFTY 50 held over the identical window (2021-02-01 to 2026-07-06, no leverage, no costs) as the passive benchmark throughout.

CAGR and maximum drawdown against the passive benchmark

Book	Return	Max DD	vs NIFTY CAGR	vs NIFTY MDD
NIFTY 50 (buy & hold)	10.39% (CAGR)	−17.01%	—	—
Model A only	20.19%	−24.06%	+9.79pp	−7.05pp
Model A + conditioned	25.00%	−17.70%	+14.61pp	−0.69pp
Model D only	17.21%	−17.51%	+6.81pp	−0.50pp
Model D + conditioned	22.02%	−13.64%	+11.63pp	+3.37pp

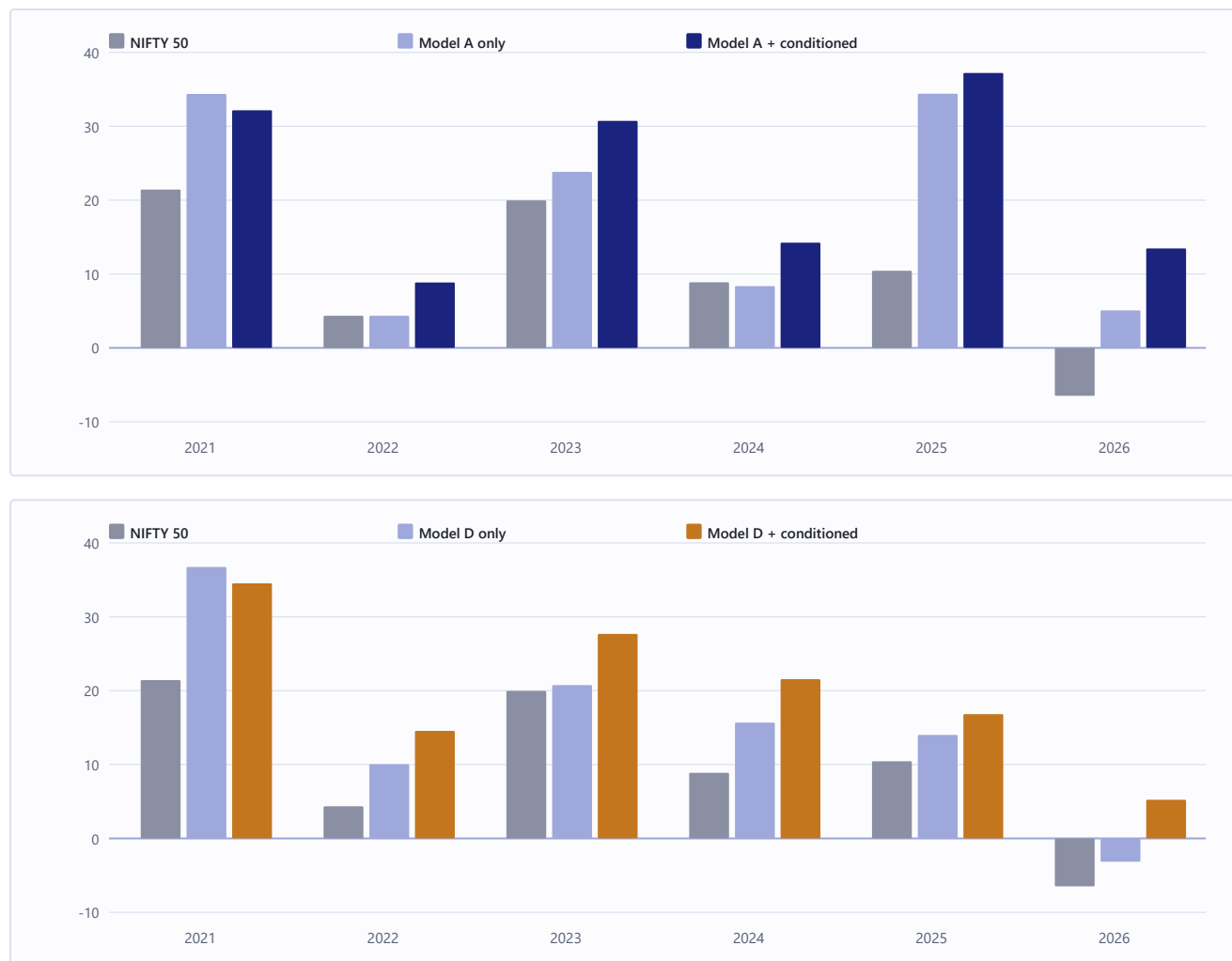
NIFTY's own return here is quoted as CAGR (the natural convention for a passive buy-and-hold), while the model rows use the simple-return mandate convention explained in §9.4 — the two are not directly additive, but both are shown so the comparison is transparent rather than convenient. **One result deserves to be stated plainly rather than buried:** Model A + conditioned still draws down slightly more (−17.70%) than simply holding the index (−17.01%) — the overlay closes most of the gap to the benchmark's own risk but does not fully close it. Model D + conditioned does close it, finishing 3.37 points better than the index on drawdown while returning more than double the index's CAGR.

8.1 Growth of Rs 100, monthly, 2021–2026



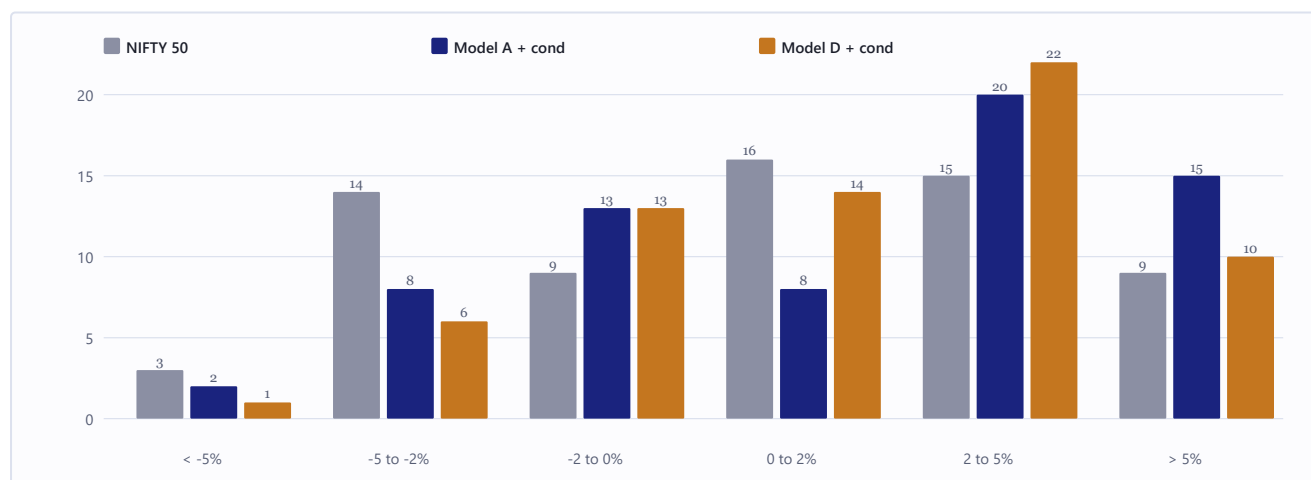
Rs 100 invested at the start of the window would have grown to Rs 168.8 in NIFTY 50, Rs 216.5 in Model A + conditioned, and Rs 198.4 in Model D + conditioned. Both books track the index shape closely (as the correlation figures in §7.1 predict) while compounding visibly ahead of it, with the gap widening most noticeably from 2024 onward.

8.2 Yearly return — each model against NIFTY 50



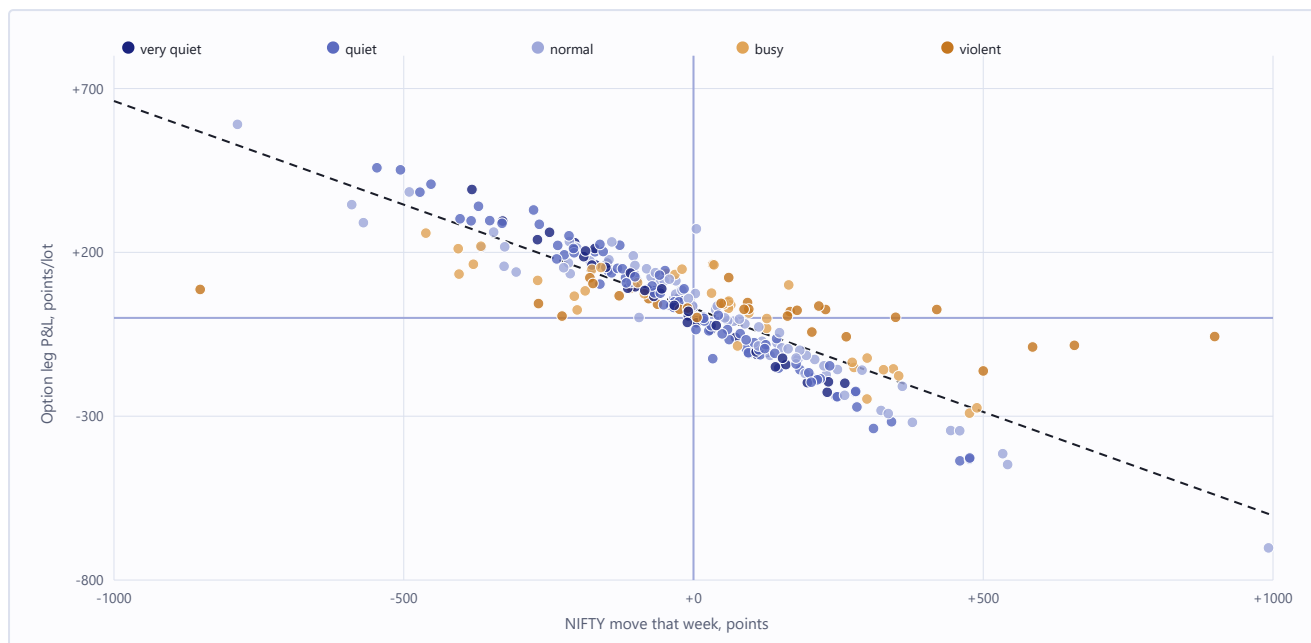
Both models beat the index in five of six calendar years measured; the conditioned overlay (darkest bar) sits at or above the "only" version (mid bar) in every year except 2021, consistent with the year-by-year table in §9.

8.3 Monthly return distribution



Both conditioned books show visibly fewer sharply negative months than NIFTY (fewer bars in the <-5% and -5%-to--2% buckets) and a heavier right tail (more months above +5%). Distribution statistics: NIFTY mean +0.89%/month, std 3.78%, skew -0.24; Model A + conditioned mean +2.07%, std 4.04%, skew +0.07; Model D + conditioned mean +1.82%, std 3.53%, skew -0.01. Model A's monthly distribution is close to symmetric (skew near zero) despite carrying the largest mean return of the three.

8.4 The hedge relationship, visualised



Every point is one traded week: the option leg's P&L in points per lot (vertical axis) against NIFTY's move that week in points (horizontal axis), coloured by the volatility regime read before the trade. The dashed line is the best linear fit across all 277 weeks. Very-quiet weeks (darkest points) cluster tightly along a steep negative line — the near-complete hedge described numerically in §7.2; violent weeks (lightest points) scatter much closer to horizontal, visibly less sensitive to the week's NIFTY move. The picture is the same regression result in §7.2, shown rather than tabulated.

9. Results

Out of sample: 2023-01-01 to 2026-07-22

Book	Return %/yr	Profit Rs	Max DD %	Sharpe	CAR/MDD	Win months %
Model A only	20.18	71,64,970	-17.05	0.89	1.18	67.4
Model A + current 2.5 σ	20.84	73,99,179	-16.87	0.94	1.24	67.4
Model A + conditioned	26.69	94,79,320	-12.17	1.55	2.19	69.8
Model D only	13.32	47,29,039	-15.75	0.50	0.85	67.4
Model D + current 2.5 σ	13.98	49,63,248	-15.47	0.55	0.90	67.4
Model D + conditioned	19.83	70,43,389	-13.64	1.15	1.45	72.1

Year by year

Return on Rs 1 crore (%). The final column is the overlay's contribution, identical for both models because lot count depends only on NIFTY spot.

Year	A only	A + 2.5 σ	A + conditioned	D only	D + 2.5 σ	D + conditioned	Overlay added
2021	34.38	34.49	32.17	36.75	36.86	34.54	-2.21
2022	4.34	4.86	8.85	10.04	10.56	14.55	+4.51
2023	23.82	23.27	30.73	20.77	20.22	27.68	+6.91
2024	8.36	10.07	14.26	15.67	17.37	21.57	+5.90
2025	34.39	35.18	37.22	14.00	14.80	16.83	+2.83
2026	5.07	5.52	13.46	-3.15	-2.70	5.24	+8.39

The overlay's contribution is **counter-cyclical to the models**: largest in their weakest years, smallest or negative in their strongest. In 2026 it converted a losing year for Model D into a positive one. This is the correct shape for a stabiliser, and it is the single most useful property of the structure.

Ten worst drawdowns

Model A + conditioned (standalone: -24.06%)

Peak	Trough	Depth %	Days
2022-04-08	2022-05-13	-17.70	115
2026-02-26	2026-03-23	-12.17	50
2022-01-12	2022-03-07	-11.86	82
2024-09-26	2024-11-21	-11.77	207
2022-09-22	2023-03-28	-10.20	246

Model D + conditioned (standalone: -17.51%)

Peak	Trough	Depth %	Days
2026-01-07	2026-03-30	-13.64	138
2022-04-13	2022-05-13	-13.45	126
2021-11-12	2021-12-20	-11.25	140
2024-12-13	2025-02-28	-9.33	131
2022-11-30	2023-02-02	-8.85	174

Every drawdown recovers; the longest recovery is 246 days.

Execution footprint

Weekly cycles / transactions	277 cycles, 744 transactions, 95 intraweek adjustments (0.34/week)
Lots held	5 to 18 (average 9.3), varying with the NSE lot size in force
Coverage of Rs 1 Cr	average 94.8%, maximum 100.0% — never naked
Margin used	average Rs 10,43,124, peak Rs 10,99,858 = 22% of the Rs 50 L pledge
Option leg P&L	Rs 26,31,511 total, Rs 4,87,317 per year, 59% of weeks profitable
Best / worst week	+Rs 2,59,452 / -Rs 2,73,624

9.4 Metric definitions, both conventions, full window

Every figure below was independently recomputed and reconciled against the trade log (§12). Where a metric admits more than one reasonable convention, both are shown; the convention used in this note is in bold and matches the house standard set in the project's existing report generators.

Model A + conditioned, 2021-02-01 to 2026-07-22 (5.47 years, 66 months, 1,351 sessions)

Metric	Convention A	Convention B
Return	Simple: 25.00%/yr (profit ÷ years ÷ Rs 1 Cr)	CAGR-equivalent: 17.07% (not used — nothing compounds under this mandate)
Volatility	Monthly × √12: 14.11%	Daily × √252: 17.35% (inflated by expiry-day settlement lumps)
Sharpe	1.35 (on monthly basis)	1.10 (on daily basis — the conservative floor)
Max drawdown	-17.70% (% of Rs 1 Cr deployed)	-12.59% (% of the running peak, NAV convention)
Profit factor	3.83 (monthly P&L)	1.29 (daily P&L)

No convention choice changes the direction of the result — conditioned beats unhedged and beats the current 2.5σ rule on every metric under either basis. The choice only affects the size of the improvement shown.

10. Validation

Approximately 4,700 configurations were searched across all sweeps against 281 weeks of data. That ratio makes curve-fitting the default explanation for any positive result, so three tests were run specifically to falsify it.

Test 1 — Walk-forward

At the start of each year, parameters were selected using only data that existed before that year, then frozen and traded for the following twelve months.

Trade year	Selected using	base	C	Option leg that year, Rs/lot
2023	< 2023 (100 weeks)	–150	50	57,212
2024	< 2024 (150 weeks)	–100	50	13,940
2025	< 2025 (202 weeks)	–100	100	1,19,238
2026	< 2026 (257 weeks)	–100	100	1,20,296

The selected parameters moved little and converged, which is the signature of a stable rule rather than a fitted one.

2023–2026 return on Rs 1 crore

Basis	Model A	Model D
No overlay	20.18	13.32
Walk-forward (no hindsight)	25.69	18.83
Full-sample best (hindsight)	26.69	19.83

About 1.0 percentage point of the headline is hindsight; about 5.5 points is genuine.

Test 2 — Placebo

The entire machine was re-run twenty times with the volatility signal randomly permuted across dates. Distribution, costs, strike-moving behaviour and search budget were all identical; only the information content was destroyed.

Measure	Value
Real signal, walk-forward gain	+Rs 24,233 / yr
Shuffled controls, mean	–Rs 9,530 / yr
Shuffled controls, best of 20	+Rs 3,455 / yr
Controls beating the real signal	0 of 20
Controls making any profit	2 of 20
z-score vs control distribution	3.55
Empirical p-value	0.048

An earlier five-shuffle run produced one control that scored close to the real signal on an in-sample best-cell basis. That comparison is uninformative, because both sides harvest selection luck from a 63-cell search. Under walk-forward the controls collapse, which is exactly the expected behaviour of noise.

Test 3 — Is the edge where the mechanism says it should be?

If the strategy works for the stated reason, essentially all of its advantage must appear in violent weeks. It does: **94% of the total edge** comes from that bucket, at Rs 3,058 per lot per week, with the other four buckets close to flat. The shuffled controls scatter their edge across buckets with large offsetting swings — the signature of luck rather than mechanism.

Test 4 — Is the linear z-score form itself justified, or arbitrary?

$K = \text{Spot} + \text{base} + C \times z$ is one of several plausible functional forms. Three competitors were tested on identical data, cost model and search budget, scored on walk-forward gain over a no-signal null:

Walk-forward gain over the no-signal null, Rs/lot/year

Form	Definition	WF gain over null
z-score (used)	$S + \text{base} + C \times z$	+22,212
Expected move	$S + \text{base} + C \times (S\sigma\sqrt{T})$	+4,888
Sigma-scaled (current 2.5σ rule's own form)	$S \times (1 + c\sigma\sqrt{T})$	-16,661

The z-score form wins by roughly $4.5\times$ over the next best, and the parameterisation used by the existing 2.5σ rule is *worse than doing nothing* once it is asked to condition on volatility rather than sit at a fixed multiple. The reason is structural: NIFTY moved from roughly 15,000 to 26,000 across the test window, so an expected-move or sigma-scaled strike grows mechanically with the index level and partly confuses "the index is higher" with "the market is more volatile." A z-score is scale-free by construction and does not have that failure mode.

The functional *form* was chosen for this reason before any fitting and survives the comparison above. The two *numbers* in it, base and C, were selected by grid search and are therefore fitted; walk-forward independently converged on base=-100, C=+100, close to but not identical with the full-sample optimum of base=-150, C=+200 used elsewhere in this note. The gap between them is the 1.0 percentage point of hindsight quantified in Test 1.

Look-ahead audit and corrections made

Realised volatility	Computed from returns ending the prior session; verified numerically against an independent recomputation.
z-score window	Mean and standard deviation taken from the 252 sessions before the trade date.
India VIX	Previous session's close.
Option quotes	Read at the same timestamp as the spot bar. 5,032,632 quotes joined with zero timestamp mismatches and zero duplicates.
Expiry mapping	281 of 281 cycles correct; no cycle expires on or before its own roll date.
Correction 1	Lot size had been held flat at 65. Replaced with the real NSE history: 75, then 50 (Jul 2021), 25 (Apr 2024), 75 (Nov 2024), 65 (2026).
Correction 2	Position size had been computed from a spot price roughly 235 points stale, leaving 13 weeks over-covered to 102.8%. Now sized from the true entry spot; coverage caps at exactly 100.0%.

Slippage stress

Model A, return on Rs 1 crore (%). Base assumption is 0.25 points per leg.

Slippage per leg (pts)	0.25	0.50	1.00	2.00	3.00
Full window	25.00	24.84	24.52	23.88	23.23
Out of sample	26.69	26.55	26.26	25.70	25.13

At twelve times the base dealing cost the structure still returns 23.23%, comfortably above the 20.74% of the current 2.5σ overlay at zero stress.

11. Risks and Limitations

What is being sold

This structure sells the right tail. In a year when NIFTY runs away to the upside it will underperform the unhedged book, as it did by 2.21 points in 2021. The steady premium is compensation for occasionally forgoing an outsized rally. That is the trade, and it should be understood as such before deployment.

Material risks

Variance premium compression	The engine is IV exceeding RV, true 69.4% of the time historically. If that gap closes the strategy stops paying. This is the single dependency.
Overnight gaps	The 300-point adjustment only operates while the market is open. A weekend or event gap passes straight through it and cannot be managed.
Basis risk	The hedge is a NIFTY index call against a portfolio of Model A or D shares. It neutralises market beta by design, preserving stock-selection alpha, but offers no protection in a week when the holdings fall and the index does not.
Deep-ITM liquidity	81% of weeks are struck in the money. At the entry bar, ATM strikes trade roughly 491,000 contracts per five-minute bar, 150-point ITM about 52,000, and 400-point ITM about 1,600 — roughly 25 lots against the 6–18 required. Executable, but the strategy is a visible part of the flow at the deep end. This is the first place real-world costs will exceed the model.
Margin on ITM shorts	In-the-money short calls attract higher SPAN than out-of-the-money. Peak usage is 22% of the pledge, leaving 4.5× headroom, but the treatment should be confirmed with the broker before sizing up.

Statistical limitations, stated plainly

- **$p = 0.048$ is one-in-twenty, not proof.** It is evidence that the signal carries information, not a guarantee that it will continue to.
- **Only about a third of the parameter grid beats doing nothing.** The working region is real but narrow. This is not a rule where approximately correct settings will suffice, which is precisely why the walk-forward result matters more than the headline.
- **The edge rests on 33 violent weeks out of 277.** That is a thin base for the component carrying almost all of the advantage, and it is the honest weak point of the analysis.
- **Model A and Model D returns are taken as given** from their existing backtests. Any survivorship or selection bias in those carries through unchanged.

12. Data and Assumptions

Period	2021-02-04 to 2026-07-22 — 281 weekly cycles, 277 traded
Option data	NIFTY five-minute option chain, 6,083,609 CE quotes across 98,404 spot bars
Spot data	One-minute NIFTY, sampled on a five-minute grid at :59 stamps
Lot size	Real NSE history — 75, 50 (Jul 2021), 25 (26 Apr 2024), 75 (20 Nov 2024), 65 (2026)
Margin model	11% of notional for a short call; pledge haircut 50% of Rs 1 crore
Costs	0.25 points slippage per leg plus 0.5% of premium turnover, covering brokerage, STT, exchange, GST and stamp
Equity cost drag	Model A 1.317% p.a., Model D 0.225% p.a., applied daily
Risk-free rate	5.92% (for Sharpe)
Settlement	Positions bought back at 15:25 on expiry; never allowed to settle, so no exercise STT

One convention note: the supplied lot-size schedule gives "Jul 2021" without a day, and 2021-07-01 has been used for the 75→50 transition. The sensitivity of the headline result to that choice is under one tenth of a percentage point.

Past results describe what did happen and are not a forecast. All figures are produced from live backtest output and reconcile to the accompanying workbook, `Conditioned_Covered_Call_Report.xlsx`.